

Co-creating nonprofit brand equity

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- *The concept of brand equity (BE) has been the subject of much recent discussion in the field of nonprofit branding and also has been the concept of brand co-creation in the field of branding. This study combines these aspects and examines the BE co-creation of a nonprofit entity, the Finnish Defence Forces military driving school. Empirical data were collected from industrial logistics service providers and military recruits. A theoretical BE model was further developed using structural equation modeling in an exploratory manner to reveal whether the BE model of the military driving school is similar from both viewpoints. Similar workable models were found, indicating strong BE co-creation and thereby contributing to the field. To support the future co-creation of BE of the military driving school, it is important to ensure that the recruits take positive experiences from the school and industrial logistics service providers carefully consider how they relay their experiences about the school to other truck drivers. The attitudes of recruits towards the military driving school are likely eventually to conform to those of civilian truck drivers. Finding workable models required omitting some BE dimensions and measures of earlier models. Moreover, using data from one country alone may be a limitation of the study. Therefore, nonprofit BE measures should be further developed and tested with data from further afield.*

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Introduction

Brands have been of interest to researchers into nonprofit organizations for decades. Although discussion and research have ranged widely, from charity brands (e.g., Sargeant *et al.*, 2007; Hudson, 2008; Mark, 2005) to co-branding (e.g., Dickinson and Barker, 2007) and re-branding (e.g., Hankinson and Lomax, 2006), brand co-creation (see, e.g., Brodie *et al.*, 2006; Boyle, 2007; Gregory, 2007; Merz *et al.*,

2009; Payne *et al.*, 2009; Juntunen *et al.*, 2012; Juntunen, 2012) in the nonprofit sector has been largely overlooked.

The fact that nonprofit organizations exhibit specific characteristics not shared by for-profit organizations has given rise to the need to explore nonprofit brand equity (BE) (Faircloth, 2005; Pope *et al.*, 2009). In general, BE has attracted much attention among researchers from both the financial (see e.g., Biel, 1992; Simon and Sullivan, 1993) and marketing (see e.g., Aaker, 1992; Keller, 1993; Berry, 2000; Davis *et al.*, 2008; Keller, 2000; Juntunen *et al.*, 2011c) viewpoints, but the viewpoint of brand co-creation has been neglected.

Here, we consider nonprofit BE from the viewpoint of brand co-creation. The aim of this paper was to develop a model of co-created BE in the nonprofit context. More specifically, the study examines

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The present study is part of a wider research project that aims to enhance co-operation between recruits of the FDF military driving school and Finnish transport companies to ensure the successful transfer of skilled and enthusiastic drivers from the military forces to transport companies, which currently suffer from the lack of a committed workforce.

whether the BE models of internal and external stakeholders of a nonprofit entity are similar. To put this in an empirical context, we have chosen the Finnish Defence Forces (FDF) military driving school. In Finland, military driving school is embedded into the recruits' military service, and military driving school is a government entity—thereby representing a nonprofit organization. Many people who work in the transport industry in Finland have been trained in the school as part of their military service and are able to pass on their experience to younger drivers and compare notes with other people in the industry. A great number of recruits in the military driving school work in the transport industry before their military service, where they form an image of the school from more experienced colleagues. After their training, the recruits often continue working in the transport industry, for example, as logistic service providers (LSPs) or truck drivers, and are similarly able to compare their experience of the school with others. Here, we consider recruits as internal stakeholders and LSPs as external stakeholders and, thereby consider the military driving school as an interesting context for studying the co-creation of nonprofit BE.

The remainder of the paper is organized as follows. First, the empirical context is described, and a literature review is presented. Thereafter, methodological issues are described, followed by results. Finally, the paper offers conclusions and suggests further topics for research.

The Finnish Defence Forces military driving school

The transport industry is an important player in services markets. Globally, 85% of domestic and 81% of international transport have been outsourced (Capgemini, 2008). According to Solakivi *et al.* (2009), in Finland, 90% of transport services have been outsourced. Further, Finland has a large number of mainly small transport companies, and hence, competition in the industry is intense, resulting in low profitability and a high rate of bankruptcies (Juntunen *et al.*, 2011b). In addition, a large number of truck drivers are approaching pension age and therefore inducing a high turnover among drivers within companies. Hence, companies

need a constant supply of trained new employees (Juntunen *et al.*, 2011b).

Basic transport driver training approved by the Finnish Vehicle Administration (Autorekisterikeskus, AKE) is offered at professional schools. However, as transport companies co-operate closely with the FDF to support both peacetime and possible wartime activities, the FDF military transport division, specifically the military driving school, has for decades provided this approved training to recruits as part of their military service, producing skilled drivers who go on to employment in civilian transport companies. Recruits who intend to make a career as drivers in the civilian transport industry may thus serve their military service in the military driving school.

Those who successfully complete their training receive a heavy goods vehicle license—and some also a public service vehicle (bus or coach) license—as well as a letter of reference, which is valued by many civilian employers. Almost all those who graduate from the military driving school continue to work as drivers after their military service. Hence, the military driving school plays an important part in the commercial environment in Finland (Juntunen *et al.*, 2011a, 2011b).

Literature review

Brand co-creation

Brand co-creation (see e.g., Brodie *et al.*, 2006; Boyle, 2007; Gregory, 2007; Fyrberg and Jürriado, 2009; Payne *et al.*, 2009; Juntunen, 2012; Juntunen *et al.*, 2012) often arises out of value co-creation (Prahalad and Ramaswamy, 2004) or co-production (e.g., Bendapudi and Leone, 2003; Etgar, 2008), and service-dominant logic, where value is seen as starting with the supplier understanding customers' value-creating processes and learning how to support their co-creation activities (Vargo and Lusch, 2004; Vargo and Lusch, 2008). Unlike product and corporate branding, the brands in brand co-creation are not built solely by organizations, but instead the customers (Boyle, 2007; Pitt *et al.*, 2006) and all other internal and external stakeholders (Brodie *et al.*, 2006; Gregory, 2007; Fyrberg and Jürriado, 2009; Payne *et al.*, 2009) who actively engage in the brand co-creation process. According to

Brodie *et al.* (2006), the service brand plays an important symbolic role in the co-creation of value by facilitating and mediating relationships for co-creating meaning and experience in the interaction of a firm's employees and stakeholders. A commonly known practical example of this are the educational services—which is also our empirical context—where the student is both the customer and the product.

The quality of interaction between network players is fundamental to co-creation (Fyrberg and Jürriado, 2009). The value-creating process can be looked at from the viewpoint of the supplier, the customer or the encounter, in which both the supplier and the customer simultaneously interact and co-create experiences (Payne *et al.*, 2008; Payne *et al.*, 2009). However, encounter experiences may vary in different phases of the relationship (Payne *et al.*, 2009).

Brand equity models

Two widely used BE models in the domain of marketing are those of Aaker (1996) and Keller (1993). Aaker's BE model consists of brand loyalty, brand awareness (BA), perceived quality, brand associations and other proprietary brand assets, whereas Keller's (1993) model focuses on brand knowledge and its two components, BA (recall and recognition) and brand image (BI) (favorability, uniqueness and strengths of brand associations, attributes, benefits and attitudes). In the context of logistics, Keller's (1993) model is frequently utilized and further developed (e.g., Davis *et al.*, 2008, Juntunen *et al.*, 2011c). Davis *et al.* (2008), examining BE in logistics service markets, proposed that BE accrues to a firm, rather than to a product. They found a positive relationship between BA and BE, as well as between BI and BE. Juntunen *et al.* (2011c) found that corporate BA and corporate BI affect corporate BE among logistics service providers. They also identified a one-way relationship between corporate BA and corporate BI.

In the nonprofit sector, Faircloth (2005) highlighted the lack of research devoted to exploring BE. Since then, however, the number of studies has increased enormously (see e.g., Stegemann and Thompson, 2005; Lee and Back, 2008; Hou *et al.*,

2009; Laidler-Kylander and Simonsin, 2009; Camarero *et al.*, 2010; Lee and Back, 2010). According to Faircloth (2005), nonprofit BE consists of brand personality (brand respect and brand differentiation), BA (brand familiarity) and BI (brand character and brand scale). Camarero *et al.* (2010) examined the differences between internal and external customers of an exhibition by considering the impact of loyalty (from previous exhibitions visited), perceived quality (various intrinsic features of the destination), BI (brand personality) and brand values (religious and cultural values of the exhibition) on BE (intention to visit future exhibitions and willingness to pay for the visit). They concluded that consumer loyalty and perceived quality are key factors in BE, regardless of visitor origin. Furthermore, BI was not a determinant of BE for internal visitors, although it was for external visitors, and the impact on BE of the brand values conveyed by the exhibition was greater on internal visitors than on external visitors.

The existing research on nonprofit BE does not explicitly use the term "brand co-creation," even though several studies seem to have elements of it. For example, the comparative study of Camarero *et al.* (2010) examines the experiences of two stakeholder groups in an encounter suggested by Payne *et al.* (2008, 2009). The study by Lee and Back (2008), which investigated conference attendee behavior from the perspective of attendee-based BE in tourism marketing, can be considered in terms of customers engaging in brand co-creation (Boyle, 2007; Pitt *et al.*, 2006). Hankinson (2000) observed that in the nonprofit sector, different stakeholders perceive brands in different ways. The importance of the various stakeholders suggested by several brand co-creation researchers (e.g., Brodie *et al.*, 2006; Gregory, 2007; Fyrberg and Jürriado, 2009) is, in a sense, a reference to brand co-creation. Thus, it is legitimate to examine nonprofit BE from the point of view of brand co-creation.

Methodology

Research model and measures

In this study, we test whether the B2B BE model of Davis *et al.* (2008) in the context of logistics can be modified and used to describe the co-creation

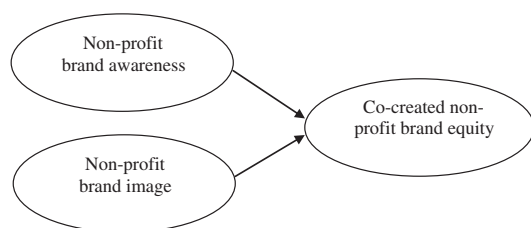


Figure 1. A model of the impacts of nonprofit brand awareness and nonprofit brand image on co-created nonprofit brand equity (modified from Davies *et al.*, 2008).

of nonprofit BE. The model was chosen because Davis *et al.* (2008) revealed that it is statistically acceptable in the context of logistics, which is our empirical context as well. The *modified* model shows whether nonprofit BA and nonprofit BI form co-created nonprofit BE (**Figure 1**). We assume that nonprofit BI and nonprofit BA are not co-created *per se*, even though nonprofit BE can be co-created.

Co-created nonprofit BE is defined here as the differential response experienced by any stakeholder to the behavior, communications and symbolism provided by or co-created with an identified nonprofit brand entity¹. It represents the different stakeholders' perspectives and refers to the value and differentiating capacity of the brand. It was measured by three questions in the questionnaire (**Table 1**). Nonprofit BA involves the stakeholders' ability to recognize and recall the brand under different conditions (Keller, 1993). It too was measured by three questions in the questionnaire. Nonprofit BI is about the stakeholders' view on the actions an entity takes and the manner in which it communicates. BI was measured by five questions in the questionnaire. Although Faircloth (2005) suggests that brand personality may have a multidimensional independent effect on BE in the nonprofit context, we follow Davis *et al.* (2008) and do not consider it as an individual factor. In the questionnaire,

¹The definition is formulated on the basis of Keller's (2000, p. 115) definition of corporate brand equity (*the differential response by consumers, customers, employees, other firms or any relevant constituency to the words, actions, communications, products or services provided by an identified corporate brand entity*) and Einwiller and Will's (2002, 101) definition of corporate branding (*a systematically planned and implemented process of creating and maintaining favorable images and consequently a favorable reputation of the company as a whole by sending signals to all stakeholders by managing behavior, communication, and symbolism*) supported by brand co-creation literature.

the respondents were asked to evaluate the military driving school. The operational measures were expressed as attitudinal statements on the basis of the 7-point Likert scale (anchored by strongly disagree and strongly agree).

Our aim was to find corresponding models for both groups. However, existing research on nonprofit BE provides certain small differences between internal and external stakeholders. Camarero *et al.* (2010) found that BI is not a determinant of BE for internal visitors, although it is for external visitors. As military driving school is also a nonprofit organization and we consider recruits as internal stakeholders and LSPs as external stakeholders, we assume that the impact of BI is higher among LSPs than recruits. Therefore, we hypothesize:

H1: The impact of BI will be greater for the LSP's than for the recruits.

Additionally, because recruits are internal stakeholders who actively engage in operating in the military driving school, we also assume that the impact of BA is higher among recruits than external LSPs, which do not currently operate within the military driving school. Thus, we hypothesize:

H2: The impact of BA is higher among recruits than LSPs.

Data collection

The empirical data consist of two data sets. The first data set, "recruits," comes from a questionnaire administered in spring 2008 to recruits of the Karelia Brigade, a fast developing Readiness Formation in eastern Finland. Because Karelia Brigade found some difficulties with online survey and they invited us to see how recruits are trained, researchers visited the driving school and similarly administer a pen and paper survey to the recruits. The theoretical sample consisted of 367 recruits who were involved in the military driving school. However, of these, 264 (71.9%) recruits were at the garrison and answered the survey when it was carried out. Of the recruits in the garrison at the moment, 97.7% were men and 2.3% women. The questionnaire consisted of various sections intended to shed light on recruits' attitudes towards the FDE. The second data set,

Table 1. Latent variables and their operational measures

Latent variable	Explanation and operational measures in the questionnaire	Label
Co-created nonprofit brand equity	Refers to equity of the nonprofit brand (measures further developed from Davis <i>et al.</i> , 2008)	BE
	We are willing to give more time to our workers to /I am willing to serve a longer military service in order to co-operate with the military driving school	Paymore
	The military driving school's brand is different from that of other driving schools	Difbrand
	The name of the military driving school gives them an advantage over other driving schools	Nameadv
Nonprofit brand awareness	Refers to awareness of the nonprofit brand (measures further developed from Davis <i>et al.</i> , 2008)	BA
	The name of the military driving school is well known in our industry/among recruits	Name
	The military driving school is recognized as a competent provider of driver training	Strong
	In comparison to other driving schools, the military driving school is the leading brand in the industry	Brand
Nonprofit brand image	Refers to image of the nonprofit brand (measures further developed from Davis <i>et al.</i> , 2008)	BI
	The military driving school is known as a driver training provider that takes good care of its students	Careful
	We can reliably predict how the military driving school will perform	Predict
	In comparison to other driving schools, the military driving school is known to consistently deliver very high quality	Quality
	In comparison to other driving schools, the military driving school is highly respected	comresp
	The military driving school is highly respected	resp

"LSPs," was gathered in 2008 from logistics service providers who were members of the Finnish Transport and Logistics association. The internet-based Webropol online survey questionnaire was sent via e-mail to 2,604 transport companies. The respondents were owners of small transport companies or people who were responsible for transport services in larger companies. Eventhough the participants took the survey in different formats, the surveys were exactly the same concerning questions relating for brand dimensions; however, there were some differences in demographic-related questions.

In the recruits' data set, we contacted the 264 who were at the garrison when survey was administered. All recruits in the room took our survey, and hence, response rate was 100%. Among recruits, 68.9% already had a truck driver's license and 26.2% had work experience in transport, although only 5.8% had had more than 1 year of such experience. In the LSP data set, questionnaire was sent to 2,604 transport companies. A reminder was sent to

those who had not answered the questionnaire in time. In total, 460 acceptable responses for the purposes of this study were returned, representing a response rate of 17.7%. Non-response bias was studied by comparing different response waves (Armstrong and Overton, 1977; Lambert and Harrington, 1990) using a randomized one-way analysis of variance. The first wave included companies that responded after the original e-mail request (69.3%) and the second wave of respondents who responded after the reminder (30.7%). No statistically significant differences (using the criterion of $p > 0.05$) between the two groups were found; therefore, non-response bias is not a problem in this study, and the sample represents the target group. Among LSPs, 59.6% had work experience in transport more than 20 years.

Estimations, made with Lisrel software (Jöreskog *et al.*, 2000; Jöreskog and Sörbom, 1993), were calculated using the maximum likelihood method on the basis of a covariance matrix.

Results

We first tested whether the data sets were statistically compatible with the model by Davis *et al.* (2008). The fit indices for the LSP model were as follows: chi-square (C-S) 366.76, degrees of freedom (DF) 41, *p*-value 0.000, root mean square error of approximation (RMSEA) 0.13, normed fit index (NFI) 0.96, non-normed fit index (NNFI) 0.95, comparative fit index (CFI) 0.97 and goodness of fit index (GFI) 0.87. The fit indices for the recruit model were as follows: C-S 146.16, DF 41, *p*-value 0.000, RMSEA 0.11, NFI 0.96, NNFI 0.96, CFI 0.97 and GFI 0.90. The chi-square tests show an unacceptable fit of the data to the models, the minimum acceptable *p*-value normally being 0.05. An RMSEA value over 0.10 would indicate a poor fit of the model (Browne and Cudeck, 1993), which was the case here. NNFI, the most important fit index for large samples, should be over 0.90 (Hu and Bentler, 1999). NFI, CFI and GFI values above 0.90 indicate acceptance (Jaccard and Wan, 1996). Because most of the fit indices suggest that the models should not be accepted, we rejected both of them.

Because we rejected both models, we did not conduct multiple group structural equation modeling (SEM) for the analysis. However, we continued by using SEM in an exploratory manner (see e.g., Kline, 2005) and reduced one operational measure both from BA (brand) and BI (predict). Modified LSPs model's fit indices were as follows: C-S 135.14, DF 24, *p*-value 0.000, RMSEA 0.11, NFI 0.98, NNFI 0.97, CFI 0.98 and GFI 0.93. The new fit indices for the recruit model were as follows: C-S 67.49, DF 24, *p*-value 0.000, RMSEA 0.085, NFI 0.97, NNFI 0.97, CFI 0.98 and GFI 0.94. With a large sample size ($N > 200$), the chi-square tests usually show an unacceptable fit of the data to the model, and hence, the NNFI is the most important fit index for large samples (Hu and Bentler, 1999). Because most of the fit indices suggest that the modified models should be accepted, we thereby managed to create structural models with adequate statistical fits for both groups (Table 2). We also checked the modification indices of Lisrel software and modified the recruits' model accordingly. By freeing the error correlation between two measures concerning respect (comresp and resp), the "recruits" model was statistically significantly improved. The fit indices

provided similar results to the previous models, indicating that this model was also acceptable.

All relationships in the models are statistically significant, and the factor loadings are good. A weak condition for convergent validity is that the factor regression coefficient on a particular item is statistically significant and a stronger condition is that the factor regression coefficient is substantial (Steenkamp and van Trijp, 1991). As the coefficients in this study are very good and statistically significant, both weak and strong conditions of convergent validity are fulfilled. Also all construct reliability values of more than 0.70 and average variance extracted values of more than 0.50 for each factor indicate that all models can be considered statistically acceptable (Table 3). In conclusion, the measures used in our models measure latent variables BA, BI and BE well. The models are quite similar in terms of factor loadings; only estimates between the factors differ between the models.

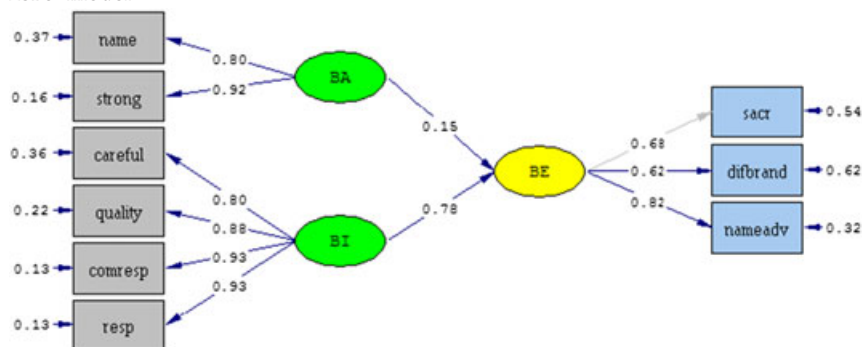
Discussion

This study concentrated on developing a model of co-created BE in the nonprofit context. More specifically, the study examined whether the BE models of internal (recruits) and external (logistics service providers) stakeholders of a nonprofit entity (military driving school) are similar. Finding workable measurement models indicates that the model of Davis *et al.* (2008) is to some extent suitable for measuring nonprofit BE. Because two almost similar models were found, we suggest that nonprofit BE can be co-created with different stakeholders.

As in previous studies (e.g., Juntunen *et al.*, 2011c), in both models, BA and BI affected BE. However, one measure from both BA and BI needed to be omitted, which is understandable in this kind of exploratory analysis. Existing models of a similar kind would not be possible by accident—there must be strong interaction between groups so identical models can exist. In fact, the majority of the LSP respondents have served their military service in the military driving school and several recruits worked as truck drivers both before and after their military service, sharing their experiences about the military driving school with others. Thus, the brand is not built solely by military forces; instead, both the recruits and the LSPs actively engage in

Table 2. Structural equation modeling models and their fit indices

LSPs' model



Chi-Square (DF) = 135.14 (24)

 p -value = 0.000

RMSEA = 0.11

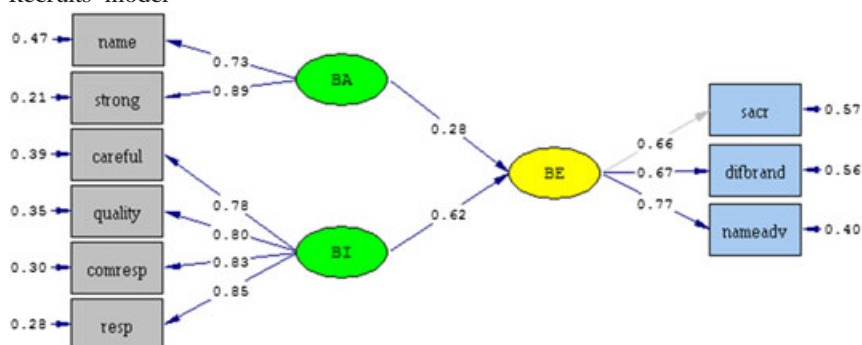
NFI = 0.98

NNFI = 0.97

CFI = 0.98

GFI = 0.93

Recruits' model



Chi-Square (DF) = 67.49 (24)

 p -value = 0.000

RMSEA = 0.085

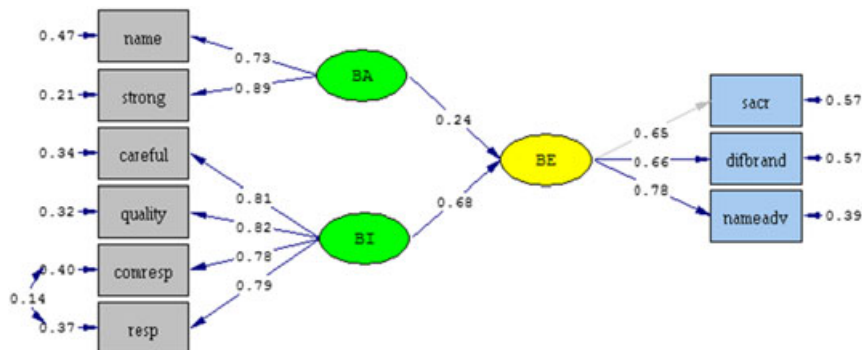
NFI = 0.97

NNFI = 0.97

CFI = 0.98

GFI = 0.94

Modified recruits' model



Chi-Square (DF) = 47.79 (23)

 p -value = 0.002

RMSEA = 0.064

NFI = 0.98

NNFI = 0.98

CFI = 0.99

GFI = 0.96

DF, degrees of freedom; RMSEA, root mean square error of approximation; NFI, normed fit index; NNFI, non-normed fit index; CFI, comparative fit index; GFI, goodness of fit index.

Table 3. Construct reliability (CR) and average variance extracted (AVE) values of the models

Model	Factor	CR	AVE	Factor	CR	AVE	Factor	CR	AVE
LSPs	BA	0.848	0.598	BI	0.937	0.759	BE	0.752	0.603
Recruits	BA	0.794	0.570	BI	0.890	0.728	BE	0.742	0.596
Recruits (mod)	BA	0.794	0.570	BI	0.876	0.719	BE	0.742	0.597

BA, brand awareness; BI, brand image; BE, brand equity.

the BE co-creation process of the military driving school. This relates to the fundamental role of internal and external stakeholders in brand co-creation (see e.g., Boyle, 2007; Brodie *et al.*, 2006; Gregory, 2007; Fyrberg and Jürriado, 2009; Payne *et al.*, 2009). Following Brodie *et al.* (2006), this is an example of how a service brand, in this case, the military driving school, has an important role in facilitating and mediating relationships for co-creating meaning and experience in the interaction of internal and external stakeholders. Thus, an interesting question is how well the military forces understand both internal and external stakeholders' value-creating processes and support their co-creation activities.

To examine differences between the models we hypothesized:

H1: The impact of BI will be greater for the LSPs than for the recruits.

H2: The impact of BA is higher among recruits than LSPs.

As we assumed, the impact of BI was greater for the LSPs than for the recruits. Thus, H1 is supported. This might be explained by remembering that the LSP group gained their experiences earlier and now have only faded recollections of their training. This is also supported by the fact that LSPs recount their experiences about the military driving school to both younger and older drivers, which increases the importance of the image. This finding differs slightly from that of Camarero *et al.* (2010), namely, that BI was not a determinant of BE for internal visitors at all, although it was for external visitors. The BA of the recruits had almost twice as much influence on BE than that of the LSP respondents. In other words, also H2 was supported. This may indicate that the recruits are more aware of the military driving school than LSPs because they are currently actively engaged with it. From the brand co-creation viewpoint, these findings may relate to the varying encounters experienced in different phases of the relationship as suggested by Payne *et al.* (2009).

Additionally, a strong error correlation between respect-related measures was found. This can be explained by noting that, although there is a large amount of transportation-related training available

in Finland, comparison between civilian and military driving schools is difficult because of the unique nature of the military driving school.

Conclusions

The study yields three theoretical implications. First, although the BE model of Davis *et al.* (2008) lacks some dimensions put forward especially for nonprofit BE (e.g., Faircloth, 2005; Camarero *et al.*, 2010; Laidler-Kylander and Simonsin, 2009; Stegemann and Thompson, 2005), it can be used for measuring nonprofit BE to some extent. Second, although BE has generated considerable research in the field of nonprofit branding (e.g., Laidler-Kylander and Simonsin, 2009; Stegemann and Thompson, 2005; Faircloth, 2005; Lee and Back, 2008; Hou *et al.*, 2009; Camarero *et al.*, 2010; Lee and Back, 2010) and although some studies (e.g., Lee and Back, 2008; Camarero *et al.*, 2010; Hankinson, 2000) have incorporated elements of brand co-creation, little attention has been paid to nonprofit branding in general or nonprofit BE particularly from the viewpoint of brand co-creation. Thus, this study offers a contribution to the field. Third, despite a wide range of brand co-creation studies (e.g., Brodie *et al.*, 2006; Boyle, 2007; Gregory, 2007; Fyrberg and Jürriado, 2009; Payne *et al.*, 2009; Juntunen, 2012; Juntunen *et al.*, 2012), examinations of BE co-creation remain scarce.

From a managerial perspective, the military forces need to understand that because most recruits will become LSPs or work for them after completing their training, their attitudes towards the military driving school will eventually conform to those of civilian truck drivers. In other words, recruits are key contributors to the co-creation of BE among internal and external stakeholders. Therefore, it is essential that the recruits take positive experiences from the military driving school so as to ensure that a positive direction in co-creation develops. To increase the number of potential truck drivers to fulfill the LSPs' future needs, LSPs will be well advised to recommend that as many young lorry drivers as possible serve their military service in the transport troops. As older truck drivers' experiences of the military driving school may influence young drivers' attitudes, LSPs should carefully think about

the kind of image they want the young drivers to form about the military driving school and communicate this image to those younger drivers.

The present study has three limitations that merit further discussion. First, the models provided lack the dimensions of brand personality (Faircloth, 2005), loyalty, perceived quality and brand values (Camarero *et al.*, 2010), among several other dimensions (e.g., Laidler-Kylander and Simonsin, 2009; Stegemann and Thompson, 2005) found to relate particularly to nonprofit BE. Second, the models work well if one measure of both BA and BI is reduced. In other words, although the remaining measures in our study appear to be valid measures for the theoretical constructs, the other measures should be further developed. Third, the data were gathered in Finland and in a particular context, namely logistics. This might limit the generalizability of the results, although the study may offer new insights for nonprofit organizations in other contexts and in other countries of the world.

The study raises some topics for further research. Although the aim of producing a workable measurement model was achieved, we recommend further research to develop more sophisticated measures of nonprofit BA and image, especially from a nonprofit brand co-creation perspective, and to validate them with systematic methods such as confirmatory factor analysis. Additionally, it would be interesting to examine if nonprofit BI and/or nonprofit BA can also be co-created. In addition, the relationship between awareness and image identified by Juntunen *et al.* (2011c) could be further examined. Finally, the research could be extended to examine brand equity co-creation in other contexts as well. We hope that this study contributes new insights into nonprofit branding and offers ideas for developing the research.

References

- Aaker DA. 1992. *Strategic Market Management*, 3rd edn. John Wiley Inc.: New York.
- Aaker DA. 1996. *Building Strong Brands*. The Bath Press: Bath, Great Britain.
- Armstrong JS, Overton TS. 1977. Estimating nonresponse bias in mail surveys. *Journal of Marketing Research* 14(3): 396–402.
- Bendapudi N, Leone RP. 2003. Psychological implications of customer participation in coproduction. *Journal of Marketing* 67: 14–28.
- Berry LL. 2000. Cultivating service brand equity. *Journal of the Academy of Marketing Science* 28(1): 128–137.
- Biel A. 1992. How brand image drives brand equity. *Journal of Advertising Research* 32(6): 6–12.
- Boyle E. 2007. A process model of brand cocreation: brand management and research implications. *The Journal of Product and Brand Management* 12(2): 79–93.
- Brodie RJ, Glynn MS, Little V. 2006. The service brand and the service-dominant logic: missing fundamental premise or the need for stronger theory? *Marketing Theory* 6(3): 363–379.
- Browne MW, Cudeck R. 1993. Alternative ways of assessing model fit. In *Testing Structural Equation Models*, Bollen K, Long JS (eds). Sage Publications Inc.: Palm Spring.
- Camarero C, Garrido MJ, Vicente E. 2010. Components of art exhibition brand equity for internal and external visitors. *Tourism Management* 31(4): 495–504.
- Capgemini. 2008. Third-party logistics. The state of logistics outsourcing.
- Davis DE, Golicic SL, Marquardt AJ. 2008. Branding a B2B service: does a brand differentiate a logistics service provider? *Industrial Marketing Management* 37(2): 218–227.
- Dickinson S, Barker A. 2007. Evaluations of branding alliances between non-profit and commercial brand partners: the transfer of affect. *International Journal of Nonprofit and Voluntary Sector Marketing* 12(1): 75–89.
- Einwiller S, Will M. 2002. Towards an integrated approach to corporate branding—an empirical study. *Corporate Communications: An International Journal* 7(2): 100–109.
- Etgar M. 2008. A descriptive model of the consumer co-production process. *Journal of the Academy of Marketing Science* 36(1): 97–108.
- Faircloth JB. 2005. Factors influencing nonprofit resource provider support decisions: applying the brand equity concept to nonprofits. *Journal of Marketing Theory & Practice* 13(3): 1–15.
- Fyrberg A, Jürriado R. 2009. What about interaction? Networks and brands as integrators within service-dominant logic. *Journal of Service Management* 20(4): 420–432.

- Gregory A. 2007. Involving stakeholders in developing corporate brands: the communication dimension. *Journal of Marketing Management* 23(1-2): 59-73.
- Hankinson P. 2000. Brand orientation in charity organisations: qualitative research into keycharity sectors. *International Journal of Nonprofit and Voluntary Sector Marketing* 5: 207-219.
- Hankinson P, Lomax W. 2006. The effects of re-branding large UK charities on staff knowledge, attitudes and behaviour. *International Journal of Nonprofit and Voluntary Sector Marketing* 11(3): 193-207.
- Hou J, Du L, Tian Z. 2009. The effects of nonprofit brand equity on individual giving intention: mediating by the self-concept of individual donor. *International Journal of Nonprofit and Voluntary Sector Marketing* 14(3): 215-229.
- Hu L, Bentler PM. 1999. Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. *Structural Equation Modeling* 6(1): 1-55.
- Hudson J. 2008. Conceptualising brand values in the charity sector: the relationship between sector, cause and organisation. *Service Industries Journal* 28(5): 615-632.
- Jaccard J, Wan CK. 1996. *Lisrel Approaches to Interaction Effect in Multiple Regression*. Sage Publications Inc.: California.
- Jöreskog KG, Sörbom D. 1993. *LISREL 8: Structural Equation Modeling with the SIMPLIS Command Language*. SSI Inc.: Lincolnwood.
- Jöreskog KG, Sörbom D, du Toit S, du Toit M. 2000. *LISREL 8: New Statistical Features*. SSI, Inc.: Lincolnwood.
- Juntunen M. 2012. Co-creating corporate brand in start-ups. *Marketing Intelligence & Planning* 30(2): 230-249.
- Juntunen J, Juntunen M, Autere V. 2011a. A case study on the role of the Finnish Defence Forces in the transport sector. *Journal of Military Studies* 2(1): 34-43.
- Juntunen J, Juntunen M, Autere V. 2011b. Protecting Finnish defence security: a logistics challenge. *International Journal of Shipping and Transport Logistics* 3(3): 295-311.
- Juntunen M, Juntunen J, Juga J. 2011c. Corporate brand equity and loyalty in B2B markets. A study among logistics service purchasers. *Journal of Brand Management* 18(4-5): 300-311.
- Juntunen M, Juntunen J, Autere V. 2012. Recruits' corporate brand co-creation experiences of the Finnish military forces. *Corporate Reputation Review* 15(2): 88-104.
- Keller KL. 1993. Conceptualizing, measuring, and managing customer-based brand equity. *Journal of Marketing* 57(January): 1-22.
- Keller KL. 2000. Building and managing corporate brand equity. In *The Expressive Organisation. Linking Identity, Reputation, and the Corporate Brand*, Schultz M, Hatch MJ, Holten Larsen M (eds). Oxford University Press: Oxford; 115-137.
- Kline RB. 2005. *Principles and Practices of Structural Equation Modeling*. The Guilford Press: New York.
- Laidler-Kylander N, Simonsin B. 2009. How international nonprofits build brand equity. *International Journal of Nonprofit and Voluntary Sector Marketing* 14: 57-69.
- Lambert DM, Harrington TC. 1990. Measuring nonresponse bias in customer service mail surveys. *Journal of Business Logistics* 11: 5-25.
- Lee J-S, Back KJ. 2008. Attendee-based brand equity. *Tourism Management* 29(2): 331-344.
- Lee J-S, Back KJ. 2010. Reexamination of attendee-based brand equity. *Tourism Management* 31(3): 395-401.
- Mark G. 2005. Building and rebuilding charity brands: the role of creative agencies. *International Journal of Nonprofit and Voluntary Sector Marketing* 10(2): 121-132.
- Merz MA, He Y, Vargo SL. 2009. The evolving brand logic: a service-dominant logic perspective. *Journal of the Academy of Marketing Science* 37: 328-344.
- Payne AF, Storbacka K, Frow P. 2008. Managing the co-creation of value. *Journal of the Academy of Marketing Science* 36: 83-96.
- Payne AF, Storbacka K, Frow P, Knox S. 2009. Co-creating brands: diagnosing and designing the relationship experience. *Journal of Business Research* 62(3): 379-389.
- Pitt LE, Watson RT, Berthon P, Wynn D, Zinkhan G. 2006. The penguin's window: corporate brands from an open-source perspective. *Journal of the Academy of Marketing Science* 34(2): 115-127.
- Pope JA, Isely ES, Asamo-Tutu F. 2009. Developing a marketing strategy for nonprofit organizations: an exploratory study. *Journal of Nonprofit & Public Sector Marketing* 21(2): 184-201.
- Prahalad CK, Ramaswamy V. 2004. Co-creating unique value with customers. *Strategy and Leadership* 32(3): 4-9.
- Sargeant A, Ford JB, Hudson J. 2007. Charity brand personality: the relationship with giving behavior. *Nonprofit and Voluntary Sector Quarterly* 37(3): 468-491.

- Simon CJ, Sullivan MW. 1993. The measurement and determinants of brand equity: a financial approach. *Marketing Science* **12**(1): 28–52.
- Solakivi T, Ojala L, Töyli J, Hälinen H-M, Lorentz H, Rantasila K, Naula T. 2009. Finland state of logistics. Publications of the Ministry of Transport and Communications 21/2009. Helsinki.
- Steenkamp JB, van Trijp H. 1991. The use of LISREL in validating marketing constructs. *International Journal of Research in Marketing* **8**(1): 283–299.
- Stegemann N, Thompson B. 2005. Visual arts marketing: the brand equity challenge facing galleries. *International Business & Economics Research Journal* **4**(12): 1–12.
- Vargo SL, Lusch RF. 2004. Evolving to a new dominant logic for marketing. *Journal of Marketing* **68**(1): 1–17.
- Vargo SL, Lusch RF. 2008. From goods to service(s): divergences and convergences of logics. *Industrial Marketing Management* **37**(3): 254–259.